SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	ITA0532	**Course Title:**	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	**Assessment:**	Assessment Tool 1 – **Scenario-Based**
Weightage:	10% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*		
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Section / Batch:	AIML/2024	**Date:**	06/08/2026

Code of Conduct

I K.Vijay Bhaskar Reddy(192425155)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Vijay Bhaskar Reddy

ASSESSMENT TOOL 2: Scenario-Based

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Scenario	**20%**					
Identification of Key Issues	**20%**					
Application of Concepts	**25%**					
Decision Making	**20%**					
Clarity of Response	**15%**					

Background Noise Removal

(a) Interpret the Scenario and Explain Noise Impact

Background noise refers to unwanted pixel variations or random disturbances present in an image. This noise may be caused by poor lighting, camera sensors, dust, or transmission errors. During image segmentation, the goal is to separate the object of interest from the background. If background noise is present, the segmentation algorithm may incorrectly identify noisy pixels as part of the object or fail to detect the actual object boundaries.

Impact of Noise:

- Produces false object regions.
- Distorts object boundaries.
- Reduces segmentation accuracy.
- Increases computational complexity.
- Leads to incorrect feature extraction and classification.

(b) Identify Key Issues Affecting Segmentation

The major issues caused by background noise are:

1. **False Edges**
   - Noise creates artificial edges that confuse edge detection algorithms.
2. **Boundary Distortion**
   - Object boundaries become irregular and difficult to detect accurately.
3. **Poor Contrast**
   - Noise reduces the difference between foreground and background.
4. **Misclassification**
   - Background pixels may be classified as foreground and vice versa.
5. **Over-segmentation or Under-segmentation**
   - Image may be divided into too many regions or important regions may merge together.

(c) Apply a Suitable Preprocessing Method

A **Median Filter** is an effective preprocessing technique for removing background noise, especially salt-and-pepper noise.

Steps:

1. Read the input image.
2. Apply a Median Filter (e.g., 3×3 or 5×5 kernel).
3. Replace each pixel with the median value of its neighboring pixels.
4. Perform image segmentation on the filtered image.

Example

Original Pixel Values

10 12 11
13 255 14
12 13 11

The value **255** is noise.

After applying the **Median Filter**:

10 12 11
13 12 14
12 13 11

The noisy pixel is replaced by the median value (**12**), while the surrounding details are preserved.

(d) Justify Your Approach Logically

The Median Filter is preferred because:

- Removes impulse (salt-and-pepper) noise effectively.
- Preserves edges better than averaging filters.
- Does not blur important image details significantly.
- Improves segmentation accuracy by reducing unwanted background variations.
- Produces a cleaner image for further processing such as thresholding, edge detection, or object recognition.

Hence, median filtering provides a good balance between noise removal and preservation of image features

(e) Conclusion

Background noise negatively affects image segmentation by introducing false edges, distorted boundaries, and incorrect object detection. Applying a **Median Filter** before segmentation effectively removes unwanted noise while preserving important image details. This preprocessing step improves segmentation accuracy, enhances object detection, and produces more reliable results for subsequent image analysis tasks.

Summary

Aspect	Explanation
Problem	Background noise affects segmentation accuracy.
Issues	False edges, poor contrast, boundary distortion, misclassification.
Preprocessing Method	Median Filtering (3×3 or 5×5 kernel).
Advantages	Removes noise, preserves edges, improves segmentation accuracy.
Result	Cleaner image with accurate segmentation and better object detection.

Python Code:

```
import cv2
import matplotlib.pyplot as plt

image = cv2.imread("input.jpg")

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

filtered = cv2.medianBlur(image_rgb, 5)

gray = cv2.cvtColor(filtered, cv2.COLOR_RGB2GRAY)

_, segmented = cv2.threshold(gray, 0, 255,
                             cv2.THRESH_BINARY + cv2.THRESH_OTSU)

plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
```

```
plt.imshow(filtered)
plt.title("After Median Filter")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(segmented, cmap="gray")
plt.title("Segmented Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```